HAOYU DONG

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EDUCATION

Department of Electronic Engineering, Tsinghua University **Beijing, China**
B.S. in Electronic Information Science and Technology Sep 2023 - Jul 2027 (Expected)
GPA: 3.8/4.0
Core Coursework:

- **Quantum & Physics:** Photonic Quantum Information | Quantum & Statistical Mechanics | Microwave & Lightwave Technology | Electrodynamics
- **Math & Computing:** Probability & Stochastic Processes | Coding Theory | Data Structures & Algorithms | Deep Learning | Advanced Calculus
- **Hardware & Systems:** Signals & Systems | Digital Logic & Processors | Electronic Circuits & Systems | Electronic System Design

ACADEMIC INTERESTS

- Quantum Communication and Quantum Key Distribution (QKD)
- Quantum Networks and Secure Communication Systems
- Systems and Hardware for Quantum Technologies

PUBLICATIONS

[1] Lixuan He, **Haoyu Dong**, Yangcheng Yu, Zhenxing Chen, Jie Feng, Xin Wang, and Yong Li, "Learning to navigate in open urban environments using a simple sim2real strategy," in *ICLR 2025 Workshop on Embodied Intelligence with Large Language Models In Open City Environment*, 2025. [Online]. Available: `https://openreview.net/forum?id=24LUDaoztW`

RESEARCH EXPERIENCE

Tsinghua University **Beijing, China**

Project: Multi-User QKD Network Based on Dispersive-Optics-Assisted Energy-Time Entanglement
Undergraduate Research Assistant, Advisor: Prof. Wei Zhang Jan 2025 - Present

- Aims to experimentally build a scalable multi-user entanglement-based QKD network translating theoretical models into deployable quantum communication infrastructure.
- Proposed a dispersive-optics-assisted energy-time entanglement based multi-user QKD architecture that supports software-defined topology reconfiguration without physical rewiring, improving scalability and deployment flexibility.
- Implemented LDPC+Cascade error correction, Toeplitz-hash privacy amplification, and GPU acceleration via CUDA kernels, sparse-matrix optimization, and Nsight profiling.
- Completed 2-, 3-, and 5-user field deployments; currently extending toward multi-entangled-source scalable architectures.

Project: Sim2Real Visual-Language Navigation in Open Urban Environments
Undergraduate Research Assistant, Advisor: Prof. Yong Li @ Future Intelligence laB (FIB) May 2024 - Apr 2025

- Developed Real-Nav, a unified vision-language navigation framework for open urban environments, enabling zero-shot Sim2Real transfer under dynamic pedestrians and lighting variations.
- Built Tsinghua-roads, a real-photo-based urban simulator using OpenStreetMap and large-scale Baidu Street View projection, and introduced social-aware decision modules and curiosity-driven pre-exploration for improved Sim2Real generalization.

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- Implemented the simulation data pipeline and ROS-based real-robot deployment, including OSM network cleaning, rasterization, street-view mapping, and multi-threaded data crawlers.
- Achieved 45.0% real-world navigation success rate, outperforming FLAME by +17%, with results reported in ICLR 2025 Workshop.

Project: Document Preprocessing and Structuring Pipeline for Domain-Specific LLMs
Undergraduate Research Assistant, Advisor: Prof. Yong Li @ Future Intelligence laB (FIB) Apr 2024 - Nov 2024

- Built a high-precision PDF-to-JSON preprocessing pipeline to clean and structure complex Chinese technical documents with tables, formulas, and mixed-language content.
- Proposed a modular layout-analysis pipeline decomposing pages into text, tables, formulas, figures, and titles, with LLM-assisted normalization for non-standard document hierarchies.
- Implemented end-to-end data collection, layout analysis and OCR integration (RapidLayout, Table-Transformer, PaddleOCR), FastAPI microservice deployment, and IoU/edit-distance based evaluation.
- Delivered a production-grade system achieving state-of-the-art accuracy on Chinese tables and formulas and adopted in an industrial LLM training pipeline.

SELECTED COURSEWORK

Coursework for *Digital Logic & Processor Fundamentals Lab* Summer 2025

- Designed a 5-stage pipelined MIPS CPU in Verilog, utilizing iverilog and GTKWave for comprehensive behavioral simulation to resolve pipeline hazards.
- Deployed the architecture on FPGA via Vivado to accelerate sparse matrix multiplication workloads, validating hardware correctness through real-time BCD visualization.

Coursework for *Coding Theory* Autumn 2025

- Developed a full-link communication system simulator implementing rate-1/2 and 1/3 convolutional codes with Viterbi decoding (hard/soft decision), analyzing bit error rate performance against SNR.
- Integrated a custom encryption module with joint source-channel coding pipelines, evaluating the trade-off between transmission security and image recovery quality under constrained channel capacity.

Coursework for *Deep Learning* Autumn 2024

- Proposed SBSPO, a self-bootstrapping alignment framework for LLaVA 1.5 that utilizes a novel Reward Union strategy to mitigate feedback length bias without external supervision.
- Achieved an 81.2% reduction in object hallucination on Object HalBench, significantly outperforming proprietary models like GPT-4V in trustworthiness benchmarks.

TECHNICAL SKILLS

Programming languages: Python, C/C++, Verilog, MATLAB.
Parallel / GPU: CUDA, Nsight.
Quantum Comms: LDPC, Cascade, Toeplitz Hash, QKD post-processing.
Robotics / Systems: ROS, Raspberry Pi, Arduino.
ML / Data: PyTorch, OpenCV, NumPy, Pandas.
Tools: Vivado, iverilog, Git, FastAPI.

ACHIEVEMENTS AND OTHERS

Volunteering:

- Recorded over 255 volunteer hours on the “Volunteer Beijing” platform.
- Served as a University Admissions Ambassador providing guidance to prospective students.
- Regular blood donor.

Personal Interests:

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- Ultimate Frisbee; Running.

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